

Yash Ajay Khandelwal

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EDUCATION

Vellore Institute of Technology

- Integrated M.Tech in Computer Science and Engineering

Sep 2022 – Jul 2027

Father Agnel Multipurpose school and Junior College

- 10th – 82.6%, 12th – 91.3%

Apr 2006 – Jul 2021

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, C#, JavaScript, TypeScript, Bash, SQL, MATLAB
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Design Patterns, Multi-threading & Concurrency, Operating Systems, Computer Networks, System Design
- **Frontend:** React, HTML/CSS
- **Backend & APIs:** Node.js, Express.js, .NET, REST APIs, FastAPI, Microservices
- **Databases:** MS SQL Server, SQLite, MongoDB, Firebase, Relational & NoSQL Databases
- **Data Processing & ML:** Pandas, NumPy, Data Validation, Data Cleaning, Feature Engineering, ONNX
- **Tools & Platforms:** Git, Linux/UNIX, AWS (EC2), Docker, CI/CD, Nginx, Agile/Scrum, Unit Testing, Code Review

EXPERIENCE

Freelance Software Developer, Pronader

Aug 2025 – Oct 2025

Tech Stack: TypeScript, Node.js, Express, SQLite, Google OAuth 2.0, YouTube Data API v3

- Built a **live chat moderation platform** using TypeScript, Node.js, Express, and the YouTube Data API v3.
- Designed **REST APIs and data workflows** for moderation rules, blacklist management, and dashboard operations.
- Implemented **asynchronous processing and automated validation** for real-time chat moderation, spam detection, and duplicate filtering.
- Developed persistent **SQLite data storage** for cross-stream blacklists and configurable moderation rules.

Freelance Software Developer, Bunny Infinite

Feb 2024 – Mar 2024

Tech Stack: C++, Windows API, PresentMon, CMake, Gemini API, Python, HTML/CSS/JavaScript

- Built a **Windows game-performance watchdog** in C++ that monitors FPS, CPU/memory utilization, and system pressure in real time to detect and diagnose performance degradation.
- Designed a **bottleneck analysis and remediation engine** combining deterministic rules with Gemini-based recommendations to auto-select safe, allow-listed corrective fixes.
- Engineered a **closed-loop optimization system** that records telemetry and remediation outcomes to learn action scores, improving fix effectiveness across sessions.
- Developed an **asynchronous local control server** with a bounded request queue and worker pool, powering REST APIs, a live telemetry dashboard, and an in-game FPS overlay.

PROJECTS

MRI-GAN based Multimodal Deepfake Detection System

[GitHub](#)

Tech Stack: EfficientNet, GANs, 1D CNN, Python [PATENT PUBLISHED]

- Architected a multimodal deepfake detection system using CNN, MRI-GAN, and temporal models.
- Implemented adversarial perceptual learning and calibrated ensemble fusion for video-level predictions.
- Trained on **23.6k DFDC videos**; validated on **Celeb-DF-v2 (6.2k)** and **FaceForensics++ (5k)** video library.
- Achieved **93% accuracy** and **0.94 ROC-AUC** on benchmark datasets.
- Optimized pipeline for CPU-only inference on sampled video frames.

HyperCarSim - End-to-End ML Data Pipeline & Simulation

[GitHub](#)

Tech Stack: C#, Python, Pandas, NumPy, TensorFlow/Keras, ONNX

- Built an **end-to-end pipeline** integrating a C# simulation, Python data processing, ML training, and ONNX runtime inference.
- Engineered automated **data ingestion, validation, and feature-engineering workflows** in Python for large-scale telemetry datasets.
- Integrated a **low-latency ML inference component**, loading trained ONNX models for real-time predictions within the simulation.
- Automated the **model training-to-deployment workflow**, converting Keras models to ONNX for direct runtime integration.

ReactionTimeMERN - F1-Inspired Multiplayer Game FullStack Website

[GitHub](#) | [Live](#)

Tech Stack: React (Vite), Node.js, Express, MongoDB Atlas, Mongoose

- Built an **F1-inspired reaction-time game** with a randomized start-light sequence and precise "GO"-to-tap timing measurement.
- Engineered **cross-device online multiplayer** with room-code lobbies, host controls, and synchronized rounds across multiple concurrent players.
- Implemented **server-side match validation**, including duplicate-join prevention and per-round attempt tracking, with results persisted to a global leaderboard and game history.
- Designed a **MongoDB-backed leaderboard** that updates each player's best time and surfaces the top 10 fastest globally.

ACHIEVEMENTS & CERTIFICATIONS

- LeetCode: [YashAjayKhandelwal](#) — Solved **250+** DSA problems, Max Rating: **1480**
- Patent Published on **Multi-Method Deepfake Detection using GAN Architecture**. [Publication Details](#)
- Microsoft Azure AI Fundamentals (AI-900) — Scored **911/1000** – [Certificate](#)
- MathWorks Certified MATLAB Associate – [Web Credential](#)
- Simulink Onramp – [Certificate](#)